

Function Point Calculator

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Introduction

Motivation:

Software estimation is critical to project planning. Function Point Analysis (FPA) is a standardized technique to estimate the size and effort of software development using user-facing functions rather than lines of code.

Why this project:

To gain hands-on experience in applying FPA principles by building a real calculator for UFP, TCF, FP, and LOC estimation.

Objectives:

- Accept inputs based on 5 function types and their complexities.
- Accept or calculate DI using 14 GSCs.
- Compute UFP, TCF, FP, and estimated LOC.

Problem Statement

The goal is to implement a console-based software tool that calculates the Function Points of a software system based on user-provided inputs:

- Functional components (EI, EO, EQ, ILF, EIF) with counts and complexities.
- Degree of Influence (manually or via GSCs).
- Final output includes UFP, TCF, FP, and LOC based on selected programming language.

Test Case

```
--- External Input ---
Enter count: 2
Select complexity:
0 = Simple, 1 = Average, 2 = Complex
Your choice (0-2): 1

--- External Output ---
Enter count: 3
Select complexity:
0 = Simple, 1 = Average, 2 = Complex
Your choice (0-2): 1

--- External Inquiry ---
Enter count: 2
Select complexity:
0 = Simple, 1 = Average, 2 = Complex
Your choice (0-2): 1

--- Internal Logical Files ---
Enter count: 1
Select complexity:
0 = Simple, 1 = Average, 2 = Complex
Your choice (0-2): 1

--- External Interface Files ---
Enter count: 2
Select complexity:
0 = Simple, 1 = Average, 2 = Complex
Your choice (0-2): 1

=> Unadjusted Function Points (UFP): 55

Would you like to enter DI manually (M) or calculate from 14 system characteristics (C)?
Your choice [M/C]: M
Enter DI manually (0-70): 30

=> Degree of Influence (DI): 30
=> Technical Complexity Factor (TCF): 0.95
=> Final Function Points (FP): 52.25

Select a programming language:
1. Assembly Language
2. C
3. COBOL/Fortran
4. Pascal
5. Ada
6. C++
7. Visual Basic
8. OO Languages
9. Smalltalk
10. PowerBuilder
11. SQL/Oracle
12. Spreadsheets
13. Graphical (icons)
Your choice: 11

=> Selected Language: SQL/Oracle
=> Average LOC per FP (AVC): 12
=> Estimated LOC: 627

=== Calculation Complete ===
```


Conclusion

This project helped us understand more about software metrics and how crucial Function Points are to evaluate a program. This project also advanced our c# skills by incorporating numerous new skills.